



Chokro

চক্রে • THE CYCLE

Doing the right thing with your waste should pay you back.

A rewards loop for correct waste disposal that only pays against verified evidence — and a marketplace where those rewards are spent with local green entrepreneurs.

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Two problems sit next to each other, and nobody solves them together.

The person who does it right

- Bangladesh throws out **25,000–30,000 tonnes** of waste a day, and only **45–50%** of urban waste is formally collected at all.
- Around **100,000 people** in Dhaka alone make a living picking through it.
- **Someone who separates their recyclables and carries them to the right bin gets nothing back. No record, no proof, no reward.**
- So the behaviour is invisible — it can't be reinforced in a person or counted in a city.

The person making something better

- Makers of handmade and recycled goods — often home-based, often women — sell through scattered social media pages.
- They compete for attention against mass-produced imports with far better distribution.
- **Two years of happy customers on one page counts for nothing on the next one. Reputation isn't portable.**
- Being handmade and low-impact is a disadvantage instead of the point.

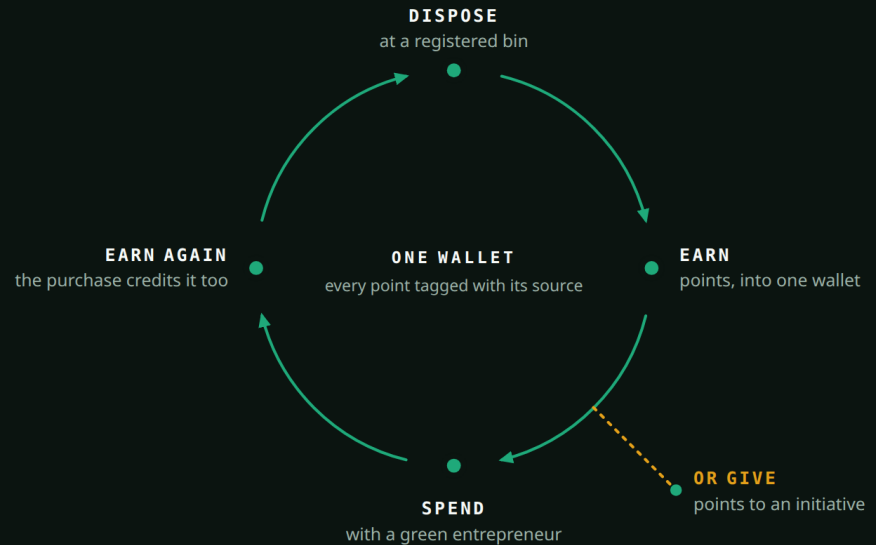
Waste apps reward disposal. Green marketplaces sell products. Neither one knows the other exists.

FIGURES: LIGHTCASTLE PARTNERS, 2025

ଚକ୍ର means cycle.

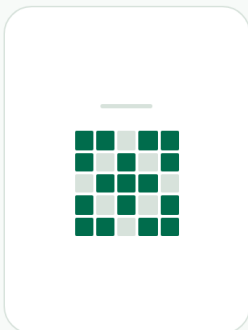
Both halves are sustainable behaviour, so **both credit the same wallet.**

The points you earn at a bin get spent on something handmade — and that purchase credits the wallet again. That is what makes this a cycle rather than a pipeline, and it is the one thing you should remember from this deck.



Four steps, standing at the bin.

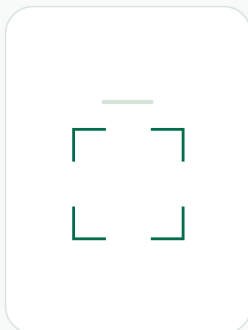
STEP 1



Scan the code on the bin

A printed sticker, not a smart bin. That is the whole reason this can actually reach a neighbourhood.

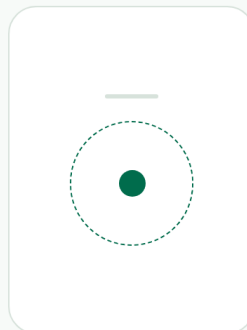
STEP 2



Photograph the disposal

And say what it is and how much. Declaring it is what gives us something to check the picture against.

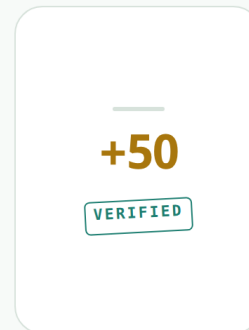
STEP 3



Your phone proves you're there

Location is captured at the moment of the scan, and checked against the bin's own recorded position.

STEP 4



Points land, with a receipt

A notification tells you the decision. Every balance change is a line in a ledger you can read back.

100 points = ₦10

One disposal is worth about ₦5

Spent in multiples of ten — the remainder stays in your wallet

A rewards system is worth exactly what its proof is worth.

Five things have to hold before a single point is paid.

WHERE

The photo has to be taken within a set radius of a registered bin. That distance is recomputed **on our server** against the bin's own recorded position — never trusted from the phone, because a phone can lie about where it is.

WHAT

An automated screen checks that the picture actually shows what the person said they were throwing away.

AGAIN

Every photo is fingerprinted, so the same picture submitted twice — at any bin, on any day — is caught.

WHEN

Six hours between claims at the same bin, and three approved disposals a day. Enough for a genuine twice-daily trip; not enough to farm.

WHO

Anything uncertain — a low confidence score, a count that disagrees with the photo, a screen we couldn't reach — goes to a person, not to a payout.

The system fails toward review, never toward payout. And no phone, ever, decides what it gets paid.

A marketplace where handmade and recycled is the premise, not the handicap.

FOR THE SHOPPER

Points come off the bill — up to half of it. They supplement money; they never replace it, so a listing always earns its seller real income.

FOR THE ENTREPRENEUR

A channel where being handmade or recycled-material is the basis of the market, plus a reputation attached to the person rather than to a page they could lose tomorrow.

FOR BOTH

Confirming the delivery is what releases the purchase points. Both sides have to show up before anything is credited.

EARN BACK

5% of every order returns as points

POINT CEILING

At most half of a bill can be paid in points

MIXED CART

Splits into one order per entrepreneur, automatically

PAYING

Cash on delivery, or a bKash / Nagad / card journey

REWARD VALUES ARE TUNABLE AT RUNTIME, NOT BAKED INTO THE APP

Or the points go to a cause instead.

Someone who would rather not shop can put what they earned straight into work on the ground.

INITIATIVE 01

Waste recovery

Getting more of what's thrown away back out of the stream.

INITIATIVE 02

Tree planting

The most legible thing a small contribution can add up to.

INITIATIVE 03

Green entrepreneurship

Backing the producers the marketplace exists for.

Choose whole points, see the effect on the balance, confirm once.

Retrying a confirmation never gives twice.

Said plainly: online giving through bKash, Nagad and card is built as a complete journey, and in this prototype it moves no real money and collects no card details. Point donations are real point movements; they are not cash and not tax-deductible.

Three kinds of people, one app, one account each.

3ZERO CHAMPION

Everyone starts here. Dispose, earn, spend, give — and read about becoming an entrepreneur when they're ready to.

3ZERO GREENPRENEUR

Still a Champion too. Lists products, manages stock, moves orders along. Applies through the app; an admin approves.

3ZERO ADMIN

Holds all three. Registers bins and prints their labels, works through whatever got flagged, tunes the reward values, suspends abuse, and watches the totals.

You don't juggle three accounts. You switch workspace — and what you're permitted to do never changes with the view you're in.

This isn't a concept. It runs.

40

FEATURES BUILT

Identity, disposal, points, marketplace, admin, self-reported actions, notifications.

2

PLATFORMS, ONE CODEBASE

Android and the browser, with no forked logic between them.

1,130

AUTOMATED TESTS

616 on the app, 282 on the service, 232 on the security rules.

41k

LINES OF PRODUCT CODE

Plus 15,500 lines of tests behind it.

The app

Flutter, one codebase

A trusted service

the only thing that can move a point

Automated photo screening

with a human fallback

Push notifications

on every decision

Printable bin labels

high error-correction QR

A ledger

every balance reconstructable

A working prototype — and the full earn-and-spend cycle runs end to end, in one continuous demo, which is the part most of these ideas never reach.

Four different people get four different things out of the same loop.

CITIES & BIN OPERATORS

Attributable, timestamped, location-verified disposal data. Which bins get used, when, and by how many distinct people — which is not measurable today at any price.

GREEN ENTREPRENEURS

A market where their materials are the premise, and a reputation that travels with them instead of living on a page.

SPONSORS

Rewards you fund are only paid against evidence that survived five checks, and every point of it is reconstructable from a ledger. You can see what your money actually bought.

THE NEIGHBOURHOOD

Money circulating between residents and small local producers, instead of leaving for mass-produced imports.

The next version of this is a pilot, not a feature.

NOW

One neighbourhood

Register a real set of bins, print and mount the labels, and run the loop with real users and a real reward pool.

NEXT

Works with no signal

A submission composed at a roadside bin should queue and send itself later. Plus a Bangla interface — the app is built for it, it isn't translated yet.

THEN

Real rails, shared data

Live payment integration in place of the simulated journeys, and disposal data handed back to the city that owns the bins.

The honest boundary: this does not solve waste management, and we won't tell you it does. It makes one link in the chain — a person's decision at the moment of disposal — measurable and worth something, and it points the resulting value at local producers instead of out of the community.

Three things, and the loop starts turning.

BINS

One neighbourhood, one operator willing to let us label their bins and record where they are.

ENTREPRENEURS

A first cohort of makers to open the marketplace with, so there is something worth spending on.

SPONSORSHIP

The reward pool and the pilot's running costs — the only place real money is needed to prove the loop.

Everything else is built.

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